



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SRI CITY

Brain Computer Interface

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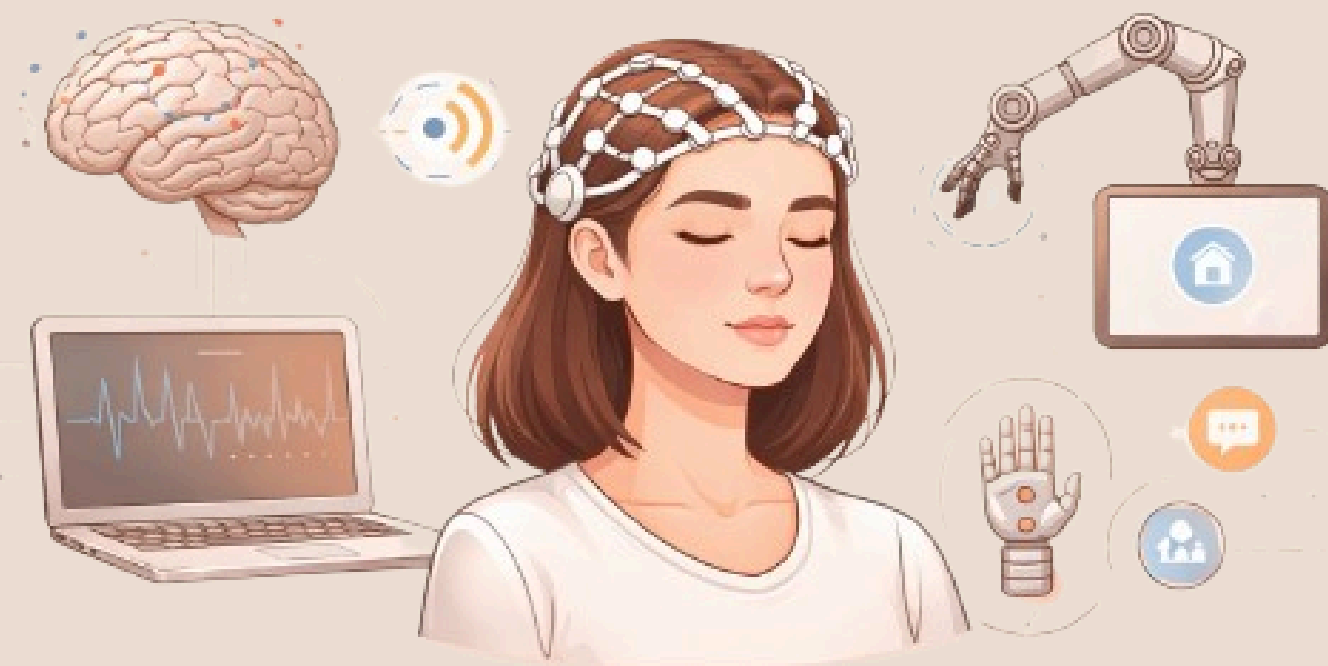
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- A Brain–Computer Interface (BCI) is a system that enables direct communication between the human brain and an external device.

BCI

Brain-Computer Interface

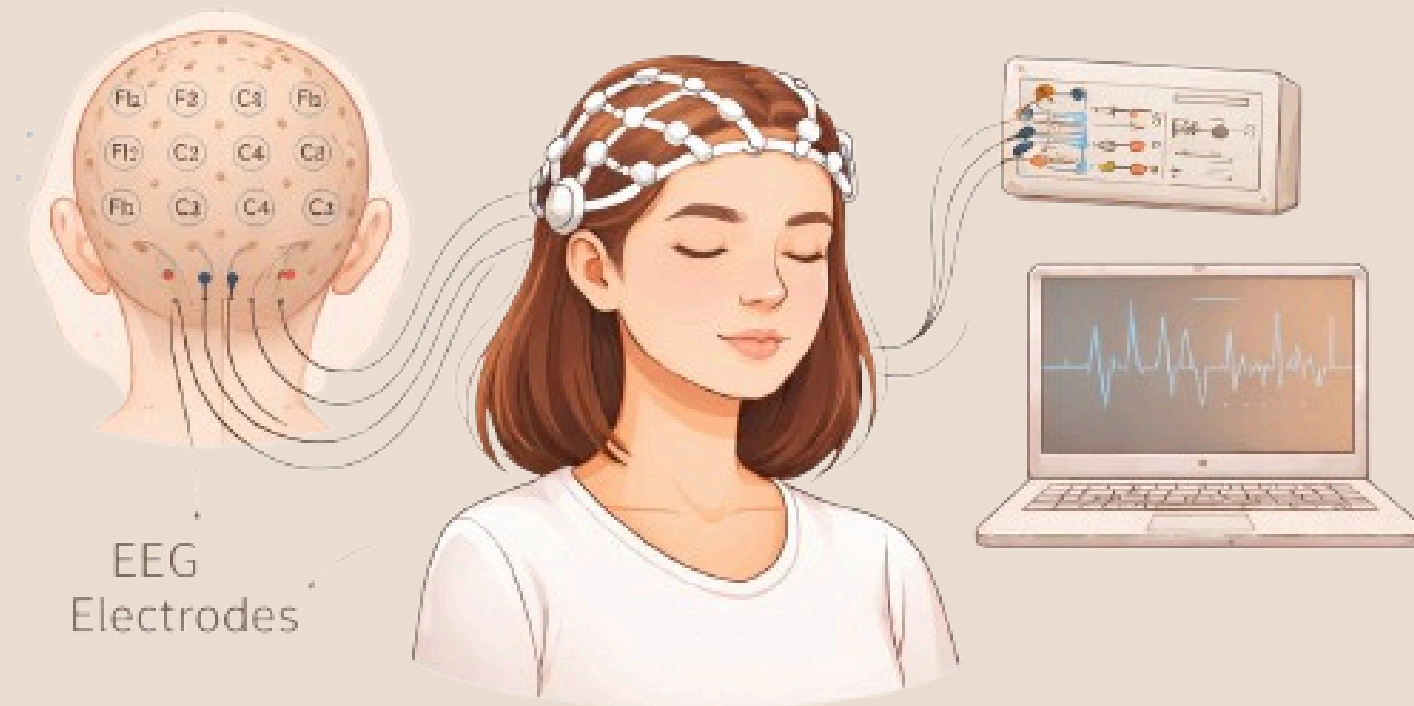


BCI have 3 main components :

- 1)Signal Acquisition:** EEG electrodes record electrical brain signals.
- 2)Signal Processing:** The signals are filtered, features are extracted, and classified.
- 3)Output Device:** The classified signals are converted into control commands for an external device.

Signal Acquisition

Signal Acquisition



- **Electroencephalography (EEG)** records brain electrical activity using scalp electrodes with conductive gel, arranged in standard systems like 10–20, 10–10, and 10–5, with the 10–10 system commonly used for accurate signal acquisition.

- **Signal acquisition refers to recording brain signals**

1. **Invasive:** Electrodes are surgically implanted into the brain's gray matter for highly precise neural recording; example: sEEG.
2. **Partially Invasive:** Electrodes are placed on the brain surface without penetrating gray matter, providing high-quality signals with lower risk; example: ECoG.
3. **Non-Invasive:** Electrodes are placed outside the skull, recording brain signals indirectly with lower quality but maximum safety; example: EEG.

Signal Processing

- **Signal Processing:** Signal processing is the process of filtering, transforming, and analyzing acquired EEG signals to remove noise and extract meaningful information for further analysis and classification.

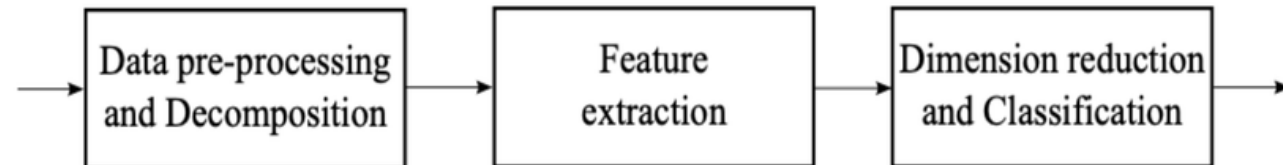
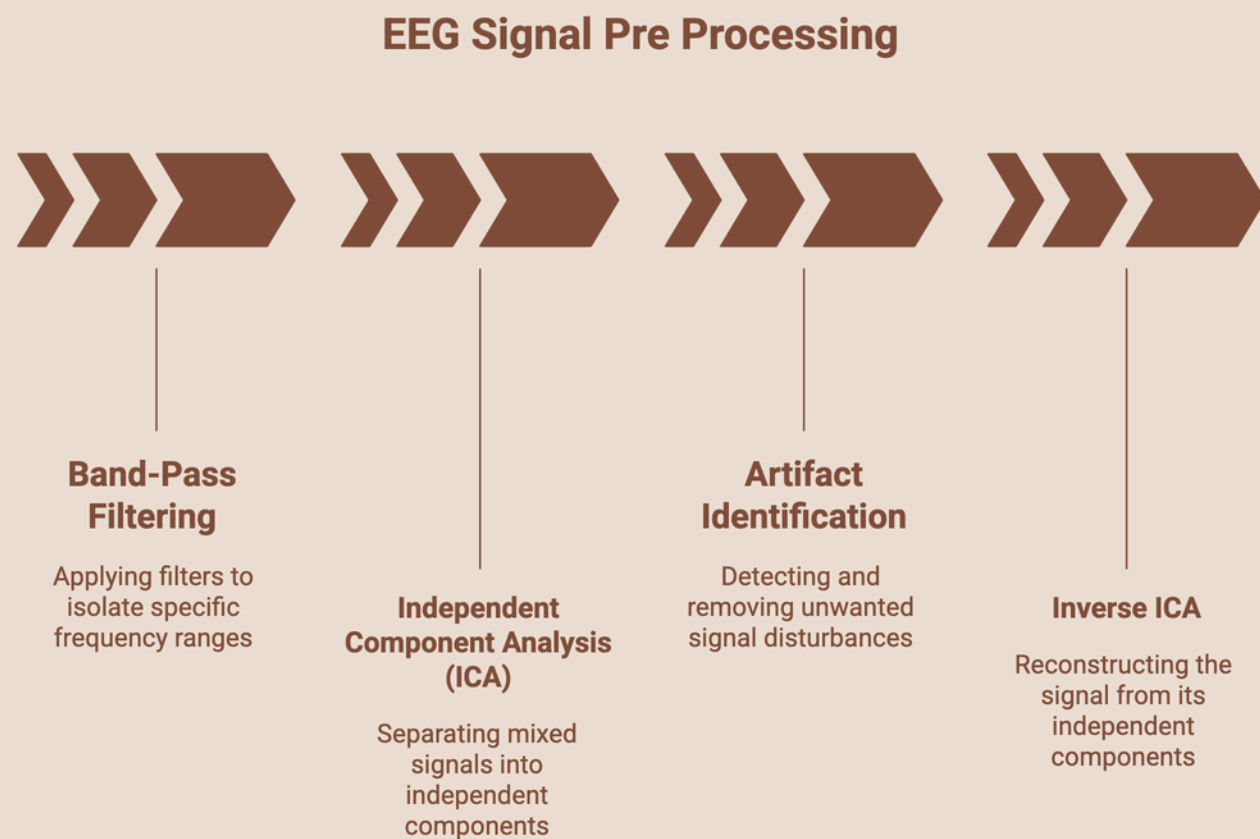
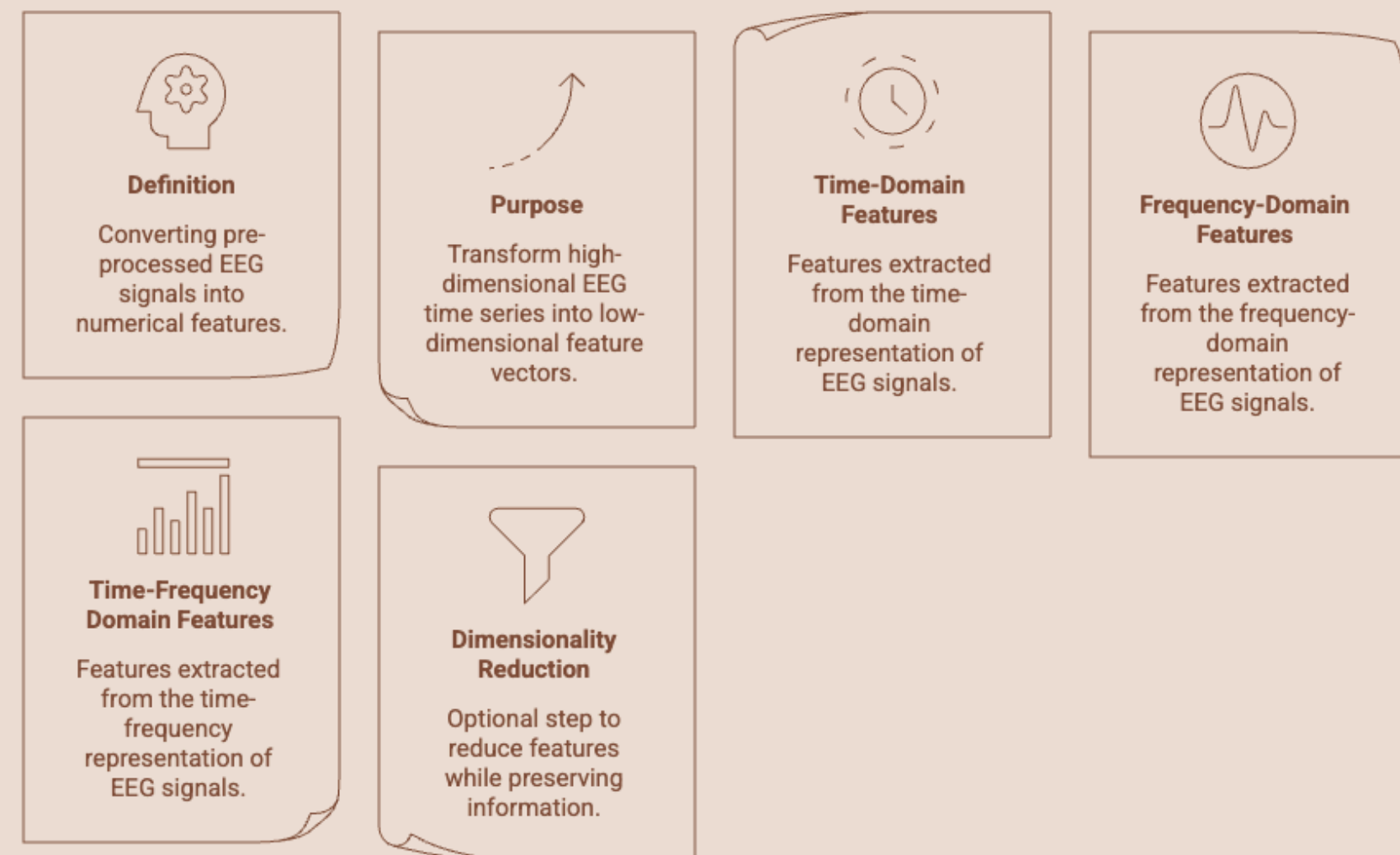


Figure. Block diagram of experimental framework. Source: [6].



EEG Feature Extraction and Dimensionality Reduction



II Stress Detection using EEG Dataset

PROBLEM STATEMENT : STRESS DETECTION USING EEG DATASETS

- Stress is primarily extracted mainly from the Prefrontal Cortex (PFC), which is responsible for decision-making, attention, working memory, and emotional regulation.
- We use the SAM40 Dataset for stress detection.

- **DATASET DETAILS:**

- 40 healthy participants
- 3 trials per participant
- Total 120 EEG segments (40 × 3)
- 32-channel EEG (10–20 system)
- Sampling rate: 128 Hz
- Duration per segment: 25 seconds
- 3200 samples per channel
- Dataset shape: (120, 32, 3200)

- **The dataset includes 4 types of induction:**

- Stroop Color–Word Test
- Mental Arithmetic
- Mirror Image Recognition
- Relaxation State

Ref :Rajdeep Ghosha,Nabamita Deba , Kaushik Senguptaa , Anurag Phukana , Nitin Choudhurya , Sreshtha Kashyapa , Souvik Phadikar b , Ramesh Sahaa , Pranesh Das c , Nidul Sinha b , Priyanka Dutta d(2021) SAM 40: Dataset of 40 subject EEG <https://doi.org/10.1016/j.dib.2021.107772>

II Stress Detection using EEG Dataset

• **Extracted EEG Features:**

- Alpha Band Power
- Beta Band Power
- Theta Band Power
- Beta/Alpha Ratio
- Entropy
- Hjorth Activity
- Hjorth Mobility
- Hjorth Complexity

Various Classifications and their Accuracy's on SAM40

Selected Features	Classifier	Relax		MI-Stress		ST-Stress		AR-Stress	
		Acc.	F-Score	Acc.	F-Score	Acc.	F-Score	Acc.	F-Score
TD-HP	SVM	99.67	99	88.5	88	87.667	87	84.17	84
FD-BP	SVM	98.17	96	88.17	88	86.5	86	84.83	84
NL-HFD	SVM	92.67	91	91.5	91	89.83	90	86.67	86
Raw EEG	EEGNet	96.67	95	93.33	93	92.5	92	91.67	91
	Shallow ConvNet	94.5	94	92.5	93	90.44	89	89.72	89
	Deep ConvNet	86.67	85	82.1	82	75.75	75	75.56	74

Table: The best accuracies in the calm and stress states are in bold : Source [\[1\]](#)

III Role of NeuroGPT-Based Model in Improving Stress Detection

Neuro-GPT Architecture for EEG-Based Stress Detection

- **Neuro-GPT Architecture:** It is a transformer-based EEG foundation model (pre-trained on large TUH EEG data) that uses an Encoder–Decoder architecture [3]
- **Pre-training Process:** Raw EEG is split into fixed segments, passed through an encoder (Convolutional + self-attention) to generate embeddings, which are then masked and predicted by a GPT decoder using reconstruction loss (MSE); the model learns general EEG patterns through backpropagation in a self-supervised manner. [3]
- **Fine-Tuning Strategy:** After pre-training on TUH EEG data, we fine-tune the model on the SAM40 dataset by dropping the GPT decoder, retaining the pre-trained encoder, and adding a fully connected classifier, resulting in an Encoder + Classifier architecture for stress detection. [5]

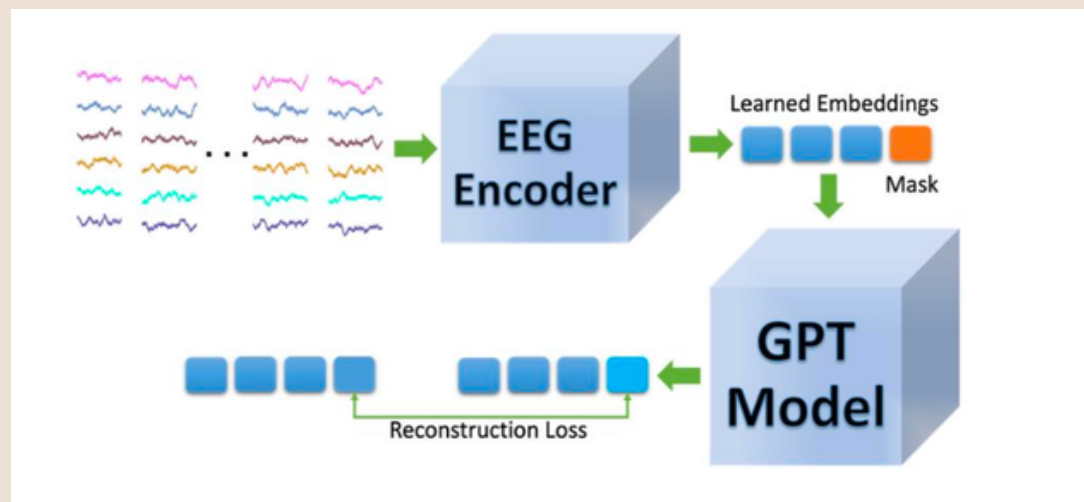


Figure. Neuro-GPT Pipeline. Source: [3].

Model	Accuracy(%)	Pros	Cons
SVM Classifier ^[1]	99.67%	• Works very well on small datasets	• Limited modeling of complex temporal patterns
Transformer Architecture ^[5]	74.40%	• Captures complex EEG patterns automatically	• Risk of overfitting on small datasets (e.g., SAM40)

V References

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